Emilio Hernandez-Martinez

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EDUCATION

University of Michigan College of Engineering August 2025 — May 2029

Bachelor's of Science in Engineering - Major in Mechanical Engineering / GPA: 3.68 *Ann Arbor, MI*

- **Relevant Coursework:** Design and Manufacturing I, General Physics I-II, Computers and Programming, Differential Equations, & Solid Mechanics
- **Additional Activities:** Michigan Research and Discovery Scholars (MRADS), FIRST Alumni & Mentors Network at Michigan (FAMNM)

Forest Hills Northern High School August 2021 — May 2025

High School Diploma / GPA: 4.50 *Grand Rapids, MI*

- **Relevant Coursework:** AP Calculus BC, AP Computer Science A, AP Computer Science Principles, AP Physics 1, AP Physics 2, & Multivariable and Vector Calculus
- **Awards:** Congressional Medal of Merit Recipient, National Merit Scholar, & Forest Hills Business Association Scholarship Recipient

EXPERIENCE

University of Michigan Quantum Engineering Laboratory September 2025 — Present

Undergraduate Researcher *Ann Arbor, MI*

- Design, fabricate, and test in-house optical hardware, including a 3D-printed fiber polarization controller.
- Perform finite-element simulations to study the physical stability of structures inside of a dilution refrigerator.
- Develop MATLAB tools to analyze simulated HBAR mode behavior and optimize resonator geometries.

University of Michigan Smart Materials & Structures Design Laboratory September 2026 — Present

Undergraduate Researcher *Ann Arbor, MI*

- Characterize force, deformation, and extension speed of pneumatically actuated robots through experimental testing.
- Evaluate mechanical behavior and object manipulation across varying module configurations and surface conditions.

FIRST Robotics Competition Team 3357 COMETS July 2021 — July 2025

Mechanical Lead, Pit Crew Member, and Robot Operator *Grand Rapids, MI*

- Managed the design and manufacturing of FRC robots, specializing in CAD and CNC operation.
- Mentored younger members in CAD software, machine shop tools, and basic robot assembly.
- Analyzed scouting data regarding competition performance and planned match strategies with other teams.

National Honor Society August 2024 — May 2025

Tutoring Coordinator *Grand Rapids, MI*

- Connected students in need of tutoring with capable NHS members and registered earned volunteering hours.
- Held open tutoring sessions twice a week and presented volunteer activities to NHS members in monthly meetings.

Grand Rapids Public Library September 2023 — April 2025

MINDSTORMS Saturdays Volunteer *Grand Rapids, MI*

- Organized an open workshop once a month throughout the school year at a local library.
- Taught elementary-age students STEM fundamentals through building and programming LEGO MINDSTORMS.

SKILLS

- **Programming Languages:** MATLAB, Python, HTML, CSS, Java, JavaScript, & Simulink
- **Software Applications:** SolidWorks, AutoCAD, Autodesk Fusion 360, Onshape, COMSOL, & Microsoft 365
- **Additional Skills:** Spanish (Native Speaker with Michigan Seal of Biliteracy)